

Laboratory # 2 ECE:4880,

Principles of ECE/CSE Design Spring 2026

Introduction

In this lab, you will design and build a frequency-selective filter that allows a desired signal to pass through while rejecting interfering signals. Specifically, we will design and build an “electric eye” light beam-interrupter safety system. The system consists of a transmitter that projects a beam of light onto a receiver (see Figure 1). When an object physically obstructs the beam, a small buzzer sounds (or a warning light is powered on). When the obstructing object is removed, the buzzer stops.

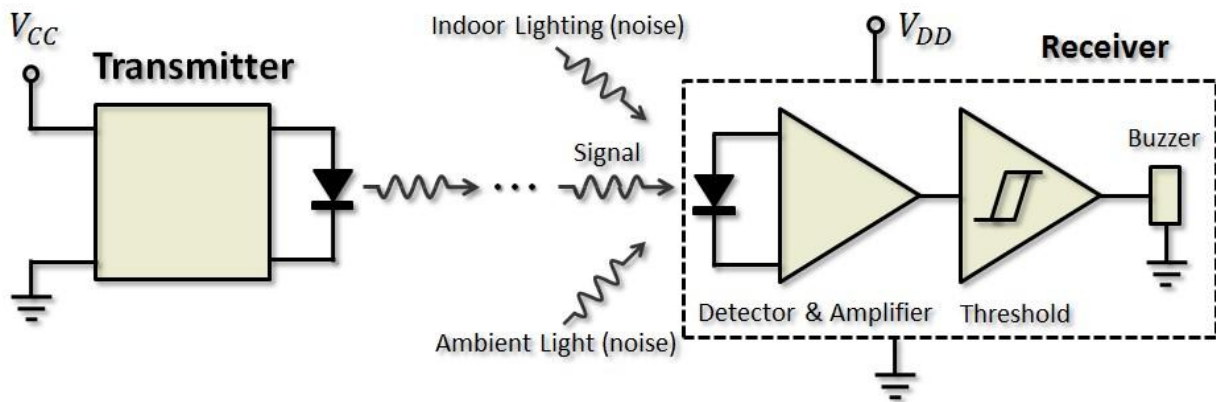


Figure 1: An “electric eye” light beam interrupter safety system.

Figures 2 (a) & (b) show example applications of electric eye safety systems. Figure 2 (a) shows a garage door closing, and the beam is interrupted by an object; the garage door stops and then opens. Beam interruption is also widely used in industry, often in the form of so-called *light curtains*. The light curtain can be a sheet of light, or several beams, that surround and act as protection for the dangerous parts of a machine, as shown in Figure 2 (b).



Figure 2 (a) “Electric eye” for a garage door opener. (b) A light curtain in an industrial setting.

The design of such a safety system requires careful consideration of several factors. For example, in a garage door application, an IR beam, invisible to humans, is preferable. However, in a factory environment, it may be preferable to have a visible light curtain.

Possible Design Approaches

In all applications of an electric eye safety system, the receiver must work reliably over a range of ambient light levels. Often, it is necessary to ensure that the system functions in the presence of artificial room lighting or nearby light sources such as desk lamps and computer screens.

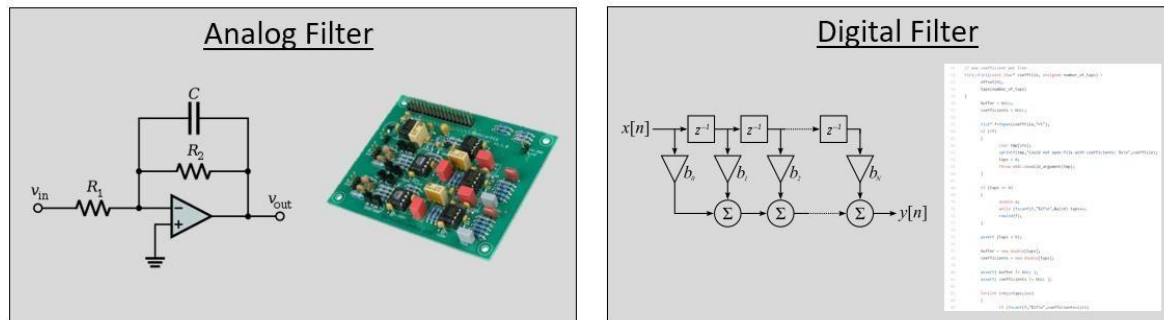
In the garage door application, the distance between transmitter and receiver, combined with misalignment, often results in quite low signal levels at the receiver. One cannot solve the problem by simply increasing the intensity of the transmitter light beam, perhaps using a laser diode, since this may become a vision hazard. Simply amplifying the signal at the receiver is not an option either: the photodetector at the receiver sees the beam but also sees fluctuations in ambient light and other light sources. If we think of the beam as the signal and the other light sources as noise, we can say that the signal-to-noise ratio (SNR) at the receiver is poor. Increasing the amplifier gain increases both the desired signal and the noise, with no improvement in the SNR; in fact, any practical amplifier will have a noise figure greater than unity, which means that the SNR is actually degraded by the process of amplification.

It should be clear from this discussion that what we need is a way of *separating* the signal of interest from the various noise signals. To facilitate this, a common technique is to gently modulate the LED at the transmitter, e.g., by using a 555-timer circuit to toggle the LED on and off, say every 2 milliseconds, and then tune the receiver to look for periodic variations in light intensity. In the above example, such a matched filter can be approximated by a *bandpass filter* tuned to 500 Hz. A bandpass filter tuned to a harmonic (i.e., a small multiple of 500 Hz) may work as well.

There are many ways to build such a filtering circuit. One simple idea is to do the filtering in the optical domain: if we choose an infrared LED at the transmitter, we can make the receiver robust to changes in ambient lighting by using a photodetector that is insensitive to the shorter wavelengths of light in the visible spectrum. However, this method is not very flexible compared to an electronic filter. The latter includes analog filters which can be simple passive circuits, or fairly complex active circuits with multiple first- or second-order sections. However, the most flexible and powerful class of filters is a digital filter, which, for relatively low-bandwidth applications like this lab, can be conveniently implemented in software on a microcontroller.

You are free to build any kind of solution you like. The specifications for this lab, however, heavily favor a digital filter, so this is what we will focus on. (Indeed, some students in the class may recognize this lab from a similar design exercise in a previous analog circuits course. However, an analog solution for this lab would require a significantly more complex design than you may expect.)

A Quick Introduction to Filters



Digital vs. Analog Filters (<https://pysdr.org/content/filters.html>)

Filters can be thought of as circuits that remember the recent history of an input voltage and average out unwanted rapid fluctuations to produce a clean, smooth output voltage. Analog filters charge and discharge capacitors (and inductors) to remember the recent history of the input, whereas a digital filter simply stores a vector of recent voltage measurements in memory to achieve the same result.

A digital filter represents analog voltages by a sequence of periodic measurements. The clock speed of a digital filter limits how fast these measurements (samples) can be made. Digital filters can only track voltages that are sufficiently slow. In this lab, the signal of interest is a periodically modulated light intensity. As a rough rule of thumb, any periodic signal can be reasonably represented digitally if we make at least 10 measurements per cycle. Thus, if the LEDs are modulated at 1 kHz, the digital filter should make measurements at least once every 100 μ s (0.0001 second 10KHz).

Once a sampling rate is chosen, a digital filter is specified by two sets of coefficients $[a_0, a_1, a_2, \dots, a_M]$, $[b_0, b_1, b_2, \dots, b_N]$, which can be thought of as the coefficients of a numerator and denominator polynomial. There are many software tools that allow you to find coefficients for a given set of frequency specifications. The FilterDesigner tool in MATLAB provides a fairly intuitive GUI interface.

Once you find a suitable set of filter coefficients, you can use the provided Arduino sketch to implement your filter in software.

Lab Requirements

A prototype of an electric eye safety system will be provided in this lab. In this system, the transmitter uses a 555 timer and an LED driver circuit to generate a modulated transmit beam. A prototype receiver that uses a microcontroller to detect changes in the output of a photodiode will also be available in Lab Room 2319 SC.

The provided prototype receiver works reliably under low levels of room lighting. Notably, it fails when the receiver is exposed to an interfering signal. Your task in this lab is to design and build a receiver that will work reliably under as many different lighting conditions as possible. In addition, we want to add a remote alert feature to the device. Specific requirements are as follows:

- Your receiver must work at least as well as the prototype receiver. Specifically, under low-light conditions, it should not have a measurably worse rate of false positives or false negatives compared to the prototype.
- Your receiver should work reliably in the presence of various light sources such as lightbulbs, computer screens, flashlights, and LEDs modulated similarly to the transmitter prototype (as long as it is at a different frequency). As noted earlier, the prototype receiver fails under this condition. A slight performance degradation (e.g., a slightly increased latency) compared to lowlight conditions is acceptable. However, the latency should not be so large as to make it unsuitable for use as a safety device.
- Every time the transmitted beam is interrupted, a notification alert should be sent to a cell phone. The notification should print a message in the format “Critical Safety Event at HH:MM XX on Month/Day/Year”, where HH is in the range 00 to 12, MM in the range 00 to 59, XX is AM or PM, Month in the range 01-12, and Day in the range 01-31. For demonstration purposes, the target for the notification can be hardcoded to a specific phone number or IP address.
- You will demonstrate your receiver using the existing prototype transmitter in the lab. You are not allowed to modify the transmitter in any way.
- The schematic for the prototype transmitter will be provided to you. Construct a replica of the transmitter for yourself. You will also have access to the prototype transmitter in the lab. You are expected to design and conduct experiments to characterize the behavior of the transmitter as needed to build a compatible receiver. Your lab report must document the design of your characterization experiments and the experiments’ observations.
- It is acceptable for your receiver to be based on the design of the prototype receiver, i.e., you can choose the design of the existing prototype as a starting point and modify it to improve performance. It is also fine to redesign the receiver from scratch.
- Your receiver should not have an abnormally large size or weight or other physical characteristics that may prevent it from being a reasonable replacement for the prototype receiver. You should be able to physically place it on the lab bench for your demo.